#01 · 8 NOV 2026 EN +60 XP



Water flow · Physics · Motion

How fast rivers carve deep rocks

NASA satellites captured stunning river loops in Utah. Discover how flowing water measures up with speed, distance, and time.

YOUR QUEST MAP

STOP 1 p.2
The story
How rivers carve canyons

STOP 2 p.3
The physics
Use v = s ÷ t for water speed

STOP 3 p.4
Your mission
Quiz, mini lab and badge

12 min read · SMP · grade 7–9 · Earn up to 60 XP

DID YOU KNOW?
These river loops cut through rock millions of years old.

STOP 1 · THE STORY

+10 XP FOR READING

The winding river of Utah

Satellite images highlighted dramatic winding river bends in Utah [1]. Water flowing across the landscape moves rock and soil grain by grain [1].



The San Juan River's goosenecks, photographed from the space station on 6 Oct 2026 [1]

NASA astronaut photograph ISS075-E-188830 (public domain)

120 m
distance measured along the river channel

20 s
time taken for water to travel the distance

6 m/s
speed of the flowing water stream

Rivers shape Earth over long periods. As water flows down slopes, gravity pulls it forward [1]. Meteorological agencies also track river and dam water levels to keep communities safe from floods [3].

Understanding how fast water moves helps scientists predict erosion and water flow [3]. Fast water carries more sand and gravel, cutting deeper canyons into stone [1].

TIMELINE

Rainfall
Water gathers in high ground and feeds river channels.

Flow
Streams rush downhill, gaining speed and carrying sediment.

Erosion
Moving water carves deep loops and canyons into bedrock.

STOP 2 · THE PHYSICS

+20 XP FOR SOLVING

Speed, distance and time

To understand how fast a river flows, we measure two things: how far the water travels and how long it takes. Speed tells us the distance covered in each second.

If distance increases, speed increases. If time increases for the same distance, speed decreases. We calculate speed by dividing distance by time.

THE KEY EQUATION

$$v = s \div t$$

Speed calculation equation

v	speed	m/s
s	distance	metre (m)
t	time	second (s)

We assume water flows at a steady average speed along the straight channel test section.

WORKED EXAMPLE

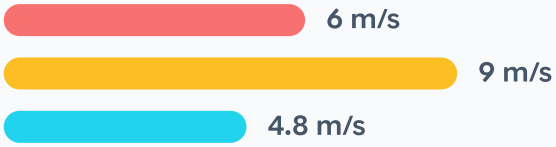
```
s = 120 m, t = 20 s
v = 120 ÷ 20
Divide: v = 6 m/s
```

v = 6 m/s

Stream	Distance (m)	Speed (m/s)
Goosenecks	120	6 m/s
Rapid	120	9 m/s
Creek	120	4.8 m/s

Different speeds over the same 120 metre river distance.

Water speed in different stream types GRAPH



Rapids flow much faster than gentle creeks over identical distances.



BULB BUDDY SAYS

Faster water flow carries heavier rocks and carves canyons deeper and faster.

STOP 3 · YOUR MISSION

+30 XP TO EARN

River flow mission

QUICK QUIZ · 3 QUESTIONS · TICK ONE BOX

Answer key: 1B · 2A · 3C

Q1 What equation calculates speed in physics? +5 XP

- A v = s × t
- B v = s ÷ t
- C v = t ÷ s
- D v = m × v

Q2 If distance s is 120 m and time t is 20 s, what is speed v? +5 XP

- A 6 m/s
- B 100 m/s
- C 2,400 m/s
- D 24 m/s

Q3 Why do fast-flowing rivers erode rock faster? +5 XP

- A They are colder
- B They contain salt
- C More energy and sediment
- D They have more sunlight

MINI LAB · TRY AT HOME OR SCHOOL

Measure stream speed at home

- [ ] Mark a 2-metre track along an inclined gutter or plastic tray with water.
- [ ] Drop a small floating leaf or cork at the start line.
- [ ] Time how many seconds it takes to reach the finish line.

Calculate your water speed using v = s ÷ t. What did you get?

BADGE UNLOCKED

Stream Tracker

You mastered river flow and speed physics.

0 / 60 XP

Erosion
When moving water wears away rock and soil over time.

Meander
A winding curve or loop in a river channel.

Speed
Distance travelled divided by time taken.

Sources

[1] NASA Earth Observatory, Taking a Gander at Utah's Goosenecks, 6 Nov 2026
[2] NASA Earth Observatory, Cyprus: Island of Snakes, Cats, and Ophiolites, 5 Nov 2026
[3] BMKG, BMKG Dorong Pemanfaatan Informasi MKG untuk Pengelolaan Bendungan, 7 Nov 2026
Cover: Jstuby / Wikimedia Commons (CC0). Story: NASA astronaut photo ISS075-E-188830 (public domain).